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S2**Started on** Thursday, 24 September 2026, 9:21 AM**State** Finished**Completed on** Thursday, 24 September 2026, 9:29 AM**Time taken** 8 mins 12 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n , find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include<stdlib.h>
3
4  int findDuplicate(int* arr, int n) {
5      int tortoise = arr[0];
6      int hare = arr[0];
7
8      do {
9          tortoise = arr[tortoise];
10         hare = arr[arr[hare]];
11     } while (tortoise != hare);
12
13     tortoise = arr[0];
14     while (tortoise != hare) {
15         tortoise = arr[tortoise];
16         hare = arr[hare];
17     }
18     return tortoise;
19 }
20
21 int main() {
22     int n;
23
24     if (scanf("%d", &n) != 1) {
25         return 1;
26     }
27
28     int* arr = (int*)malloc(n * sizeof(int));
29     if (arr == NULL) {
30         return 1;
31     }
32
33     for (int i = 0; i < n; i++) {
34         if (scanf("%d", &arr[i]) != 1) {
35             free(arr);
36             return 1;
37         }
38     }
39
40     int duplicate = findDuplicate(arr, n);
41     printf("%d\n", duplicate);
42
43     free(arr);
44
45     return 0;
46 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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